

From Continuous Geometry to Discrete Lattices: An Empirical Anatomy of Failed and Corrected Information-Geometric Models of Twin Primes

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Abstract

We document a structural formalization of information-geometric models for twin prime distributions. An initial framework (RAPF v2.6–v3.4) modeled prime gaps using a continuous hyperbolic surface of revolution with constant negative curvature, predicting a universal scale $\lambda \approx 3.70$ and a Fisher information metric $g = 1/(C_2\lambda^2)$. This construction failed when tested against validated sieve data: the continuous distribution assumption ignores that twin gaps are strictly constrained to a discrete modulo-6 arithmetic lattice. Furthermore, we demonstrate that while comparing empirical data to the local Hardy–Littlewood endpoint prediction $(\log X)^2/(2C_2)$ produces an apparent 12% discrepancy, evaluating the correct cumulative mean gap $\lambda_{\text{cum}}(X) = X/(2C_2 \text{li}_2(X))$ matches the empirical sieve data to within 0.01% at $X = 10^8$, meaning the classical Hardy–Littlewood main term requires no discrete correction at computable scales.

Recognizing that the classical continuous model is predictively highly accurate but geometrically inappropriate for discrete arithmetic objects, we construct a formalized discrete information-geometric model on the modulo-6 lattice. We derive the exact discrete Fisher information metric $I(\lambda) = 9/[\lambda^4 \sinh^2(3/\lambda)]$, and prove that it correctly converges to the continuous exponential Fisher information $1/\lambda^2$ as $\lambda \rightarrow \infty$. We further establish algebraically that no local prime arithmetic obstruction can force the twin prime density to structural zero in any twin-admissible residue class. This work illustrates the necessity of respecting discrete lattice constraints in prime modeling, and constitutes Paper 1 in a three-part research program (the post-v4.3 RAPF trilogy), establishing the discrete geometric foundation complemented by the equidistribution census of Paper 2 and the triplet extension of Paper 3.

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1 Introduction

The twin prime conjecture — that there are infinitely many pairs $(p, p + 2)$ both prime — remains one of the most famous open problems in number theory. The Hardy–Littlewood k -tuples conjecture provides the standard asymptotic prediction $\pi_2(X) \sim 2C_2 \int_2^X dt/(\log t)^2$ with $C_2 \approx 0.66016$. Recent years have seen various attempts to approach prime distributions through information geometry, differential geometry, and statistical physics.

This paper documents a concrete case study in the construction, testing, failure, and reconstruction of such a model. We begin with RAPF (Recursive Admissibility Prime Framework) versions v2.6 through v3.4, which posited that twin prime gaps could be modeled on a continuous hyperbolic surface with constant negative curvature. We present our empirical testing methodology and the

results that falsified the model’s quantitative claims, having identified the root causes of failure. A key finding is that comparing empirical data to the local Hardy–Littlewood endpoint prediction $(\log X)^2/(2C_2)$ produces an apparent $\sim 12\%$ discrepancy, whereas the correct cumulative comparator $\lambda_{\text{cum}}(X) = X/(2C_2 \text{li}_2(X))$ matches the empirical sieve data to within 0.01% at $X = 10^8$ — confirming that the classical main term is predictively accurate but geometrically inappropriate for discrete arithmetic objects.

This paper is Paper 1 in a three-part research program (the post-v4.3 RAPF trilogy):

- **Paper 1 (this work):** The discrete information-geometric framework establishing the lattice structure and Fisher information metric for twin primes.
- Paper 2: The equidistribution census providing six primorial scales ($n = 3$ through $n = 8$).
- Paper 3: Extension to prime triplets on the modulo-30 lattice, with the exact Fisher metric $I(\lambda) = 225/[\lambda^4 \sinh^2(15/\lambda)]$ and an equidistribution census through $n = 7$.

Relationship to companion papers. The discrete Fisher information metric and lattice structure derived here supply the structural foundation for the equidistribution census of Paper 2. The algebraic non-vanishing analysis in Paper 1 establishes admissibility at all scales, while the empirical results of Paper 2 demonstrate that twin-admissible classes are uniformly occupied across six primorial scales. Together, Papers 1 and 2 form a complete architecture: the discrete geometry defines the structure, and the census confirms the equidistribution. This framework is subsequently extended to higher-order dimensions via the modulo-30 triplet lattice in Paper 3, which demonstrates that the sub-Poisson boundary effect generalizes to prime triplets.

2 The Original Framework: RAPF v2.6–v3.4

The prior framework constructed a Riemannian manifold $\mathcal{M}_3$ as a surface of revolution with line element:

$$ds^2 = dr^2 + \lambda^2 e^{-2r/\lambda} d\theta^2, \quad (1)$$

yielding constant negative Gaussian curvature:

$$K = -\frac{f''(r)}{f(r)} = -\frac{1}{\lambda^2}. \quad (2)$$

The model made several quantitative claims:

Claim 1 (Static Scale): $\lambda \approx 3.70 \pm 0.01$ is a universal constant, derived from minimization of an entropy functional $S(r) = -\frac{\gamma}{2} \log(r/\lambda) + \alpha e^{-\beta r/\lambda}$ with phenomenological boundary parameters $\alpha = 1.08$, $\beta = 2.7$.

Claim 2 (Exponential Distribution): Prime gap distances r follow a continuous exponential distribution $P(r) = \frac{1}{\lambda} e^{-r/\lambda}$.

Claim 3 (Fisher Metric Scaling): Conditioning on the admissible subset defined by the twin prime singular series scales the Fisher metric as $g_A = 1/(C_2 \lambda^2)$, yielding a “34% information surplus” $\Delta g \approx 0.0376$.

Claim 4 (Spectral Correspondence): The manifold’s negative curvature enforces Jacobi field divergence consistent with GUE spectral statistics, bridging prime gaps to Riemann zero spacings.

Claim 5 (Geometric Obstruction): The Fisher information cannot vanish at any finite scale, providing an “information-theoretic obstruction” to the finiteness of twin primes.

No empirical validation had been performed.

3 Empirical Falsification

3.1 Sieve Implementation

We implemented a segmented prime sieve over $[2, 10^8]$ in Python. The sieve divides the range into 1 MB segments, sieves each segment using base primes up to $\sqrt{10^8} = 10,000$, and collects all primes. The implementation was validated against the accepted value $\pi_2(10^8) = 440,312$.

Prior Bug Identified: (An earlier implementation with cross-boundary defects was corrected; the present count has been independently verified against standard tables [8].)

3.2 Observed Results vs. Model Predictions

From the 440,312 twin prime pairs, we extracted 440,311 twin-to-twin gaps (distances between consecutive twin pairs).

Result 1 (Scale Parameter): The observed mean gap is $\hat{\lambda} = 227.11$. The model predicted $\lambda = 3.70$. The ratio is 61.4. This is not a statistical fluctuation but a structural error.

Result 2 (Distribution Shape): A Kolmogorov–Smirnov test of the normalized gap distances against the continuous exponential distribution $\text{Exp}(1)$ yields $D = 0.0506$, $p < 10^{-10}$. The continuous exponential hypothesis is decisively rejected. An Anderson–Darling test confirms the rejection with $A^2 = 1310.2$, far exceeding the 5% critical value of 1.341.

Result 3 (Multimodality): The gap distribution is strongly multimodal with peaks at 6, 12, 18, 24, 30, ... (all multiples of 6). This reflects the arithmetic constraint that every prime $p > 3$ satisfies $p \equiv 1$ or $5 \pmod{6}$, and for $(p, p+2)$ to be a twin pair, $p \equiv 5 \pmod{6}$. The next twin pair q also satisfies $q \equiv 5 \pmod{6}$, so $q - p \equiv 0 \pmod{6}$. No continuous model can capture this discrete lattice structure.

Result 4 (Fisher Metric): The observed Fisher metric from the data is $g_{\text{obs}} = 1/(227.11)^2 \approx 1.94 \times 10^{-5}$. The model’s claimed value (even with corrected scaling direction) was on the order of 10^{-2} . Three orders of magnitude of discrepancy.

Result 5 (Scale Dependence): The Hardy–Littlewood asymptotic predicts a local endpoint density $\lambda_{\text{HL}}(X) = (\log X)^2/(2C_2)$. At $X = 10^8$, this gives $\lambda_{\text{HL}} \approx 257.00$, while the observed cumulative mean gap is 227.11 — an apparent 12% discrepancy. However, comparing a cumulative mean to a local endpoint density is a category error: the correct cumulative comparator $\lambda_{\text{cum}}(X) = X/(2C_2 \text{li}_2(X))$ evaluates to 227.08 at $X = 10^8$, agreeing with observation to 0.01%. The 12% gap is simply the first-order correction term $2/\log X$ from the asymptotic expansion $\text{li}_2(X) \approx X/(\log X)^2 + 2X/(\log X)^3$. This confirms λ is a dynamic, scale-dependent parameter, not a universal constant.

4 Root Cause Analysis

The model failed for three fundamental reasons:

4.1 Continuous vs. Discrete Distribution

The continuous exponential PDF $P(r) = \frac{1}{\lambda}e^{-r/\lambda}$ assigns probability to all $r \in \mathbb{R}^+$, but twin-to-twin gaps exist exclusively on the arithmetic lattice $\{6k : k \in \mathbb{N}\}$. A smooth density function applied to a discrete lattice is structurally incorrect at the definitional level, not merely approximately wrong.

4.2 Static vs. Dynamic Scale Parameter

The derivation of $\lambda \approx 3.70$ from the entropy functional $S(r)$ treated λ as a fixed geometric constant. In reality, the mean gap between twin primes grows with X according to the Hardy–Littlewood density $\lambda(X) \sim (\log X)^2/(2C_2)$. The model’s entropy functional does not incorporate this scale dependence, making its variational minimization meaningless.

4.3 Fisher Conditioning Error

The Fisher information derivation claimed that conditioning on admissible classes scales $g \rightarrow 1/(C_2\lambda^2)$. The correct Poisson thinning analysis shows that multiplying event density by $C_2 < 1$ increases the mean gap ($\lambda \rightarrow \lambda/C_2$) and correspondingly decreases the Fisher information. The “information surplus” was an artifact of a sign error in the scaling direction.

5 The Reconstruction: Discrete Information Geometry on the 6-Lattice

Having identified all failure modes, we construct a corrected model from first principles.

5.1 The Modulo-6 Lattice PMF

Twin-to-twin gaps g take values in $\mathcal{L}_6 = \{6k : k = 1, 2, 3, \dots\}$. We model the lattice index k with a geometric distribution having an exponential envelope:

$$P(k; \lambda) = \left(e^{6/\lambda} - 1\right) e^{-6k/\lambda}, \quad k \in \{1, 2, 3, \dots\}. \quad (3)$$

The normalization is verified by the geometric series $\sum_{k=1}^{\infty} e^{-6k/\lambda} = e^{-6/\lambda}/(1 - e^{-6/\lambda})$. The distribution is a geometric PMF with success parameter:

$$p = 1 - e^{-6/\lambda}. \quad (4)$$

The mean and variance of k are:

$$\mathbb{E}[k] = \frac{1}{p} = \frac{e^{6/\lambda}}{e^{6/\lambda} - 1}, \quad \text{Var}(k) = \frac{1-p}{p^2} = \frac{e^{-6/\lambda}}{(1 - e^{-6/\lambda})^2}. \quad (5)$$

5.2 Exact Fisher Information

The log-likelihood is $\log P(k; \lambda) = \log(e^{6/\lambda} - 1) - 6k/\lambda$, with score function:

$$\frac{\partial}{\partial \lambda} \log P(k; \lambda) = \frac{6}{\lambda^2} (k - \mathbb{E}[k]). \quad (6)$$

The Fisher information is:

$$I(\lambda) = \mathbb{E} \left[\left(\frac{\partial}{\partial \lambda} \log P(k; \lambda) \right)^2 \right] = \frac{36}{\lambda^4} \text{Var}(k) = \frac{36}{\lambda^4} \cdot \frac{e^{-6/\lambda}}{(1 - e^{-6/\lambda})^2}. \quad (7)$$

Using the identity $4 \sinh^2(x) = e^{2x} - 2 + e^{-2x}$ with $x = 3/\lambda$, we obtain the closed form:

Theorem 1 (Discrete Fisher Information). *For the geometric PMF on the modulo-6 lattice $\mathcal{L}_6$ with scale parameter $\lambda > 0$, the Fisher information is*

$$I(\lambda) = \frac{9}{\lambda^4 \sinh^2(3/\lambda)}. \quad (8)$$

Asymptotic limit: For $\lambda \gg 6$, $\sinh(3/\lambda) \approx 3/\lambda$, so $I(\lambda) \approx 1/\lambda^2$, recovering the continuous exponential Fisher information. At $\lambda = 227.11$, the ratio $I_{\text{discrete}}/I_{\text{cont}} = 0.99994$.

Remark 1 (Asymptotic Behavior and the Discrete–Continuous Transition). *Because $\lambda(X) \sim (\log X)^2/2C_2$ diverges as $X \rightarrow \infty$, the discrete lattice structure asymptotically vanishes: the spacing 6 becomes negligible relative to $\lambda(X)$, and the discrete Fisher information converges to its continuous counterpart $1/\lambda^2$. The primary importance of the discrete formulation is therefore not a permanent structural separation from continuous models, but rather the stabilization of the information-geometric framework at low-to-medium computational scales (up to $\sim 10^{10}$), where the 6-lattice constraint is significant. At sufficiently large X , the continuous approximation becomes increasingly accurate, and the discrete model’s advantage is a finite-size correction rather than an asymptotic distinction.*

5.3 Dynamic Scale and Renormalization Flow

Hardy–Littlewood density gives the renormalization flow:

$$\lambda(X) = \frac{(\log X)^2}{2C_2}. \quad (9)$$

As X increases, $\lambda(X)$ diverges logarithmically. The Fisher information evaluated along this flow is:

$$I(\lambda(X)) = \frac{9}{\lambda(X)^4 \sinh^2(3/\lambda(X))} \sim \frac{4C_2^2}{(\log X)^4}. \quad (10)$$

The information decays polynomially in $1/\log X$ but remains strictly positive for all finite X .

6 Empirical Validation of the Discrete Model

6.1 Quantitative Agreement

Remark 2 (Continuous vs. Discrete Approximation). *At $X = 10^8$, the observed mean gap is $\hat{\lambda} = 227.11$. The discrete Fisher information evaluates to $I(227.11) = 1.939 \times 10^{-5}$. The continuous approximation $1/\lambda^2$ yields the identical value to four significant digits. The mathematical discrepancy between the two models is strictly $O((3/\lambda)^2)$, derived from the second-order expansion of the sinh function, confirming that the continuous model is asymptotically equivalent but geometrically imprecise at highly constrained local scales.*

6.2 Goodness-of-Fit

Testing the discrete geometric PMF against lattice indices $k_i = g_i/6$:

- KS test: $D = 0.0506$, $p < 10^{-10}$ — the simple geometric model is rejected
- Anderson–Darling: $A^2 = 1310.2$ — rejected at 5% level

This rejection is expected. The single-parameter geometric PMF captures the exponential envelope but does not account for secondary modular correlations (e.g., multiplicative constraints mod 30, 210) that create fine structure in the gap distribution. The model is *structurally* correct (discrete 6-lattice) but *phenomenologically* incomplete. This distributional rejection does not invalidate the macroscopic utility of the framework: because the Fisher information metric functions as an integrated second-moment operator, it effectively averages over these localized fine-structure perturbations, capturing the global scale information robustly while remaining invariant to local modular noise. A formal proof that $\sum_k M(k) \cdot k^2 e^{-6k/\lambda}$ vanishes in its contribution to the second moment remains an open problem.

Remark 3 (Higher-Order Structure and Density Fluctuations). *The multimodal peaks at $k = 1, 2, 3, \dots$ (gaps 6, 12, 18, $\dots$) show unequal heights that the geometric PMF predicts as a pure exponential decline. This rejection ($p < 10^{-10}$) reflects deeper statistical phenomena than just higher-order modular constraints. **Maier’s Theorem** [2] proves that prime density in short intervals exhibits significant oscillations (“clumping”), violating strict Poissonian assumptions. Empirical work by Brent [3] indicates that twin prime distributions are statistically “more random” than the base prime distribution, suggesting that the admissibility constraints partially dampen Maier-style fluctuations. The rejection of the pure geometric model reflects real higher-order structure—Maier-style density oscillations and modular constraints at mod-30, mod-210, etc.—that the single-parameter exponential envelope does not capture. The discrete Fisher information $I(\lambda)$, as an integrated second-moment functional, averages over these localized perturbations, but the fine-structure deviations remain visible at the level of goodness-of-fit.*

6.3 Scale Dependence Verification

The $\log^2 X$ scaling of $\lambda(X)$ can be tested by comparing sieve results across multiple decades. The discrete model predicts:

- $\lambda(10^6) \approx 144.6$
- $\lambda(10^7) \approx 196.8$
- $\lambda(10^8) \approx 257.0$
- $\lambda(10^9) \approx 325.3$

The cumulative mean gap $\hat{\lambda} = X/\pi_2(X)$ must be compared against the cumulative Hardy–Littlewood prediction $\lambda_{\text{cum}}(X) = X/(2C_2 \text{li}_2(X))$ rather than the endpoint-local $\lambda_{\text{HL}}(X) = (\log X)^2/(2C_2)$. Table 1 shows both comparators. The convergence to $\hat{\lambda}/\lambda_{\text{cum}} \rightarrow 1$ confirms the $\log^2 X$ scaling driven by the renormalization flow $\lambda(X) = (\log X)^2/(2C_2)$.

Table 1: Cumulative mean gap $\hat{\lambda} = X/\pi_2(X)$ vs. cumulative Hardy–Littlewood prediction $\lambda_{\text{cum}}(X) = X/(2C_2 \text{li}_2(X))$ across four decades. The ratio converges rapidly to 1.

X	$\pi_2(X)$	$\hat{\lambda} = X/\pi_2(X)$	$\lambda_{\text{cum}} = X/(2C_2 \text{li}_2(X))$	Ratio
10^6	8,169	122.41	121.24	1.010
10^7	58,980	169.55	170.20	0.996
10^8	440,312	227.11	227.08	1.000
10^9	3,424,506	292.01	291.94	1.000

The convergence ratio $\hat{\lambda}/\lambda_{\text{cum}}$ approaches 1 from both directions: $1.010 \rightarrow 0.996 \rightarrow 1.000 \rightarrow 1.000$, confirming the Hardy–Littlewood cumulative prediction to within 0.01–0.02% at computable scales.

7 Asymptotic Structure: Non-Vanishing Information

The renormalization flow implies:

$$\lim_{X \rightarrow \infty} \lambda(X) = \infty, \quad \lim_{X \rightarrow \infty} I(\lambda(X)) = 0. \quad (11)$$

However, for any finite X , we have $I(\lambda(X)) > 0$. The Fisher information decays as $(\log X)^{-4}$, which is extremely slow. At $X = 10^{15}$ (well beyond any feasible sieve), $I \approx 1.22 \times 10^{-6}$ — still strictly positive and statistically measurable.

Observation 1. *The discrete information-geometric model demonstrates that the statistical structure of twin primes is permanent and scale-invariant at all computable scales. There is no finite threshold at which the information content collapses to zero. Any proof of finiteness would need to identify a mechanism for information collapse at some unobserved $R_{\max}$ that is not reflected in the Hardy–Littlewood density or the geometric PMF structure.*

This is the corrected version of Claim 5 from the original framework. The new claim is statistical, not deductive: the information does not vanish at any finite scale, and the geometric model provides no mechanism for information collapse. This does not constitute a proof of infinitude; it establishes the absence of a finite-scale collapse mechanism within the information-geometric framework.

8 Algebraic Non-Vanishing: Local Obstruction Analysis

Having established that the discrete information-geometric model carries strictly positive Fisher information at all finite scales, we now address a complementary question: does any local prime arithmetic obstruction exist that could force the twin prime density to structural zero?

8.1 The Twin-Admissible Condition

Fix a primorial modulus $P_k = \prod_{i=1}^k p_i$. A residue class $a \pmod{P_k}$ is called *twin-admissible* if $\gcd(a, P_k) = 1$ and $\gcd(a + 2, P_k) = 1$, i.e., neither a nor $a + 2$ is divisible by any prime $p \leq p_k$. The number of such classes is

$$A_k = \prod_{2 < p \leq p_k} (p - 2). \quad (12)$$

8.2 The Heuristic Weight Product

Under the Hardy–Littlewood independence heuristic, the structural weight for a twin pair $(n, n + 2)$ in class $a \pmod{P_k}$ factors as a product over all primes:

$$\mathcal{P}_a = \prod_p \left(1 - \frac{\nu_p}{p} \right), \quad (13)$$

where ν_p is the number of solutions to $x(x + 2) \equiv 0 \pmod{p}$. We have $\nu_2 = 1$ (since $n, n + 2$ share parity) and $\nu_p = 2$ for $p > 2$ (roots $0, -2$ are distinct).

Conditioning on the admissible class $a \pmod{P_k}$, the local factors for $p \leq p_k$ become deterministic:

- For $p \leq p_k$: since $a \not\equiv 0$ and $a + 2 \not\equiv 0 \pmod{p}$ by admissibility, the local weight is 1 (no obstruction).

- For $p > p_k$: the residue of $n \pmod{p}$ is unconstrained by $n \equiv a \pmod{P_k}$, and the local weight remains $(1 - 2/p)$.

Thus for any twin-admissible class:

$$\mathcal{P}_a = 1 \cdot \prod_{p > p_k} \left(1 - \frac{2}{p}\right). \quad (14)$$

8.3 Structural Non-Vanishing

Theorem 2 (No Local Obstruction). *For any twin-admissible residue class $a \pmod{P_k}$ with $P_k \geq 2$, there exists no prime p such that the local factor $(1 - \nu_p/p)$ vanishes. The heuristic weight $\mathcal{P}_a$ contains no zero factors.*

Proof. For $p \leq p_k$, admissibility guarantees the local weight is 1, which never vanishes. For $p > p_k$, the local weight is $(1 - 2/p)$. Setting $(1 - 2/p) = 0$ gives $p = 2$, but since $P_k \geq 2$, all primes in the range $p > p_k$ satisfy $p \geq 3 > 2$. Hence $(1 - 2/p) > 0$ for all $p > p_k$. No factor in the product vanishes. $\square$

Remark 4. *Theorem 2 establishes that the admissibility condition is both necessary and sufficient to eliminate local prime obstructions. The “density door” is never structurally closed — no single prime can forbid twin primes from an admissible class. However, this is not a proof of infinitude: the infinite product $\prod_{p > p_k} (1 - 2/p)$ converges to 0 by Mertens’ theorem, reflecting the global thinning of prime density. The gap between “no local obstruction” and “infinitely many twins” is precisely the parity barrier identified by Selberg. Theorem 2 establishes that the admissibility condition is the precise algebraic criterion for a residue class to remain a candidate for twin primes.*

9 Future Directions

1. **Extended sieve validation:** The validated sieve has been completed through 10^9 , confirming $\pi_2(10^9) = 3,424,506$ with $\hat{\lambda} = 292.01$. The cumulative convergence ratio evaluates to $\hat{\lambda}/\lambda_{\text{cum}} = 1.000$, validating the scale invariance of the cumulative Hardy–Littlewood prediction up to the billion scale.
2. **Refined PMF with Maier Modulation:** Develop a multi-parameter lattice PMF $P(k) = M(k) \cdot (e^{6/\lambda} - 1)e^{-6k/\lambda}$ where $M(k)$ captures Maier-style density oscillations alongside modular correlations from mod-30 and mod-210 sieve layers. Following Brent (2014), future work will quantify exactly how the twin-prime sieving operation dampens these local fluctuations compared to the base prime distribution.
3. **Cross-tuple comparison:** Already accomplished in Paper 3, which extends the discrete Fisher framework to prime triplets on the modulo-30 lattice. The census confirms that sub-Poisson variance suppression generalizes: at small primorial scales, the $\text{CV}/\text{CV}_{\text{Poisson}}$ ratio is furthest from 1.0 (41–46% suppression at $n = 3, 4$), decreasing toward the Poisson limit as $n \rightarrow \infty$ (6.5% suppression at $n = 7$ across 17,920 classes), with mirror-symmetric chiral behavior observed between the $(0, 2, 6)$ and $(0, 4, 6)$ patterns.
4. **Information-theoretic bounds:** The Cramér–Rao lower bound states that any unbiased estimator $\hat{\lambda}$ of the scale parameter based on N observations satisfies $\text{Var}(\hat{\lambda}) \geq 1/(N \cdot I(\lambda(X)))$. At $X = 10^9$, with $N = 3,424,506$ observed twin pairs, the single-observation variance is

$1/I(\lambda) \approx 85,276$. This gives an overall estimator variance of $\text{Var}(\tilde{\lambda}) \geq 0.0249$, corresponding to a highly precise standard deviation floor of $\sigma \geq 0.16$. This means the intrinsic information structure of the modulo-6 lattice permits extreme macroscopic precision, even though the uncertainty of any single gap remains bounded by the single-observation floor ($\sigma \approx 292$), which perfectly accommodates local Maier-style fluctuations.

10 Conclusion

This paper documents a complete cycle of model construction, empirical falsification, and reconstruction. The original continuous information-geometric framework (RAPF v2.6–v3.4) failed every quantitative test against validated sieve data, with errors ranging from 6000% in the scale prediction to complete structural mismatch between continuous distributions and discrete arithmetic lattices. The failures were traced to three root causes: ignoring the modulo-6 constraint, treating a dynamic parameter as constant, and an algebraic sign error in the Fisher conditioning argument.

The corrected discrete model on the modulo-6 lattice derives exact closed-form Fisher information $I(\lambda) = 9/[\lambda^4 \sinh^2(3/\lambda)]$ and identifies $\lambda(X)$ as a renormalization flow driven by Hardy–Littlewood density. For $\lambda \gg 6$, the discrete Fisher information converges to its continuous limit $1/\lambda^2$, with the deviation governed by the second-order term in the sinh expansion. The simple geometric PMF is nevertheless rejected by goodness-of-fit tests due to unmodeled higher-order modular structure. The asymptotic analysis shows that Fisher information never vanishes at any finite scale, establishing a permanent statistical structure in the twin prime sieve.

Complementing this statistical permanence, our local obstruction analysis proves algebraically that no single prime can force the twin density to structural zero within any twin-admissible residue class (Theorem 2), confirming that the arithmetic pathway remains permanently open.

This discrete geometric foundation is complemented by the equidistribution census of Paper 2, which confirms uniform prime distribution across twin-admissible classes at primordial moduli through $n = 8$.

The discrete information-geometric approach formalized here establishes the structural necessity of lattice constraints in k -tuple modeling.

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